

BHOJPURI ASR: ZERO-SHOT VS. TRAINED MODELS BENCHMARK REPORT

Empirical Word Error Rate (WER) Evaluation, Dataset Rows, and Feature Specifications

1. Executive Summary & Evaluation Overview

This evaluation report presents a comprehensive comparison of **Zero-Shot ASR ability** versus **Locally Trained and Fine-Tuned Models** on a target Bhojpuri evaluation dataset (data/merged_bhojpuri/eval). A total of **1,054 audio test rows** comprising both clean studio speech (610 rows) and crowdsourced rural mobile recordings (444 rows) were evaluated using **Word Error Rate (WER)**.

2. Official Model Benchmark, Dataset Rows & WER Results Table

Model Name	Training Paradigm	Dataset Rows & Duration	Trainable Params	Final WER (%) ↓	Best Checkpoint	Status / Remarks
Whisper-Small (Zero-Shot)	Zero-Shot Baseline (Untrained)	0 Rows (0 Hours) Pre-trained: 680k h web	0 (0%)	~71.50% – 75.00%	—	Baseline reference out-of-the-box
Vakyansh Wav2Vec 2.0	Acoustic Model (CTC, bhom_60)	~36,000 Rows (~60.0 Hours)	~95M (100%)	108.22%	102.96% Studio 114.88% Field	High WER due to domain mismatch & CTC insertion errors
Whisper-Small (Full FT)	Full Parameter Fine-Tuning	103,746 Rows (~150.2 Hours)	~242.6M (100%)	40.58%	40.58% (Step 14,900)	Strong linguistic adaptation on 103,746 training rows
■ Whisper-Small + LoRA v1	PEFT / LoRA (r=8, q+v)	103,746 Rows (~150.2 Hours)	884.7K (0.36%)	38.91%	38.86% (Step 1,800)	Best overall accuracy; +1.67% improvement over full FT
Whisper-Small + LoRA v2	PEFT / LoRA (r=16, q+k+v+out)	103,746 Rows (~150.2 Hours)	~1.77M (0.73%)	38.91%	38.88% (Step 600)	Tied v1 accuracy; early convergence at step 600

3. Dataset Schema, Columns & Evaluation Split Breakdown

The training and benchmark datasets contain the following metadata columns and feature schemas:

Dataset Corpus	Row Count / Duration	Columns / Feature Schema (Fields)	Target Domain & Script
Whisper Training Split (AI4Bharat + SYSPIN)	103,746 Rows (~150.2 Hours audio)	6 Columns: audio (16kHz array), file (wav path), transcript (Devanagari text), duration (sec), speaker_id, topic_domain	6 Topic Domains (Agriculture, Health, Politics, Finance, Food, General) in Devanagari
Vakyansh Pre-training (ULCA / SYSPIN)	~36,000 Rows (~60.0 Hours audio)	3 Columns: audio_filepath (16kHz 1D waveform), duration (sec), text (65 Devanagari character tokens)	Pure character-level Devanagari vocabulary (65 tokens, no language model)
Merged Benchmark Eval Split (data/merged_bhojpuri/eval)	1,054 Rows (~4.8 GB Arrow cache)	610 Studio Rows (IISc SYSPIN) + 444 Mobile Rows (AI4Bharat Rural Women) Columns: audio, file, transcript, normalized_text	Ground-truth normalized Bhojpuri Devanagari test split

4. Key Scientific Insights & Conclusion

- **Impact of 103,746 Training Rows:** Moving from 0 rows (Zero-Shot ~75% WER) to 103,746 rows reduced WER to **40.58%** via Full Fine-Tuning.
- **LoRA Parameter & Sample Efficiency:** LoRA achieved the lowest project WER of **38.91%** while updating only **884,736 parameters (0.36%)** on the exact same 103,746 dataset rows.
- **CTC vs. Autoregressive Model Divergence:** The Vakyansh CTC model (trained on ~36,000 rows) scored 108.22% WER due to heavy character insertion errors on noisy mobile speech.